Phraseology

CM 1 : Welcome and Introduction

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1 Before we start...

2 Introduction

3 Multiword Expressions

4 Multiword Expressions and NLP

About me



You will see me in...

- Prompt Engineering M2
- Software Projects M2
- Written Corpora M2

About this course

7 lectures

- Phraseology and Multiword expressions
- Fixation, Defixation and Refixation
- Multiword expressions and NLP

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- Oral session (50 %)
- Final exam (50 %)

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Final exam: December 8th

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What is Phraseology ?

[1] https://en.wikipedia.org/wiki/Phraseology

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Wikipedia Definition[1]

> Phraseology is the study of set or fixed expressions, such as idioms, phrasal verbs, and other types of multi-word lexical units (often collectively referred to as phrasemes), in which the component parts of the expression take on a meaning more specific than, or otherwise not predictable from, the sum of their meanings when used independently.

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Phraseology studies fixed sequences of words.
Several words, but only one meaning.

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A bit of History

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Phraseology is a young discipline

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- Term coined by Charles Bally in 1909
- since 1970 : "phraseological boom" (Granger and Meunier, 2008)

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Phraseology is at a crossroad between several disciplines

- Phraseology **is not** just a subdomain of syntax
- Phraseology **is not** just a subdomain of lexicology
- Phraseology **is** a central domain of language sciences (Dostie and Tutin, 2025)

Terminological Nightmare

clause, conventional collocation, colligation, discourse structuring device, formulae idiom, irreversible binomial, phrase, prefab, quotation, simile, ...

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- Incompatible properties (Gross, 1996)

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For any given classification, there will always be objects that fall outside of it (Heinz, 1993).

Conventional names

- Idiom
- Phraseme
- Set phrase
- Fixed expression
- Multiword expression

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During this class, we will mainly use the term "Multiword expression" (MWE).

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How to define a MWE ?

[2] "locutions qui ont le caractère d'unités plutôt que de séries"

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Bally (1909) definition

"phrases that function as units rather than as series"[2]

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- the lexical and syntactic flexibility of its components
- the semantic role of its components and its level of compositionnality

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Exercise Time !

Study the MWEs within the following sentences

- Max kicked the bucket
- Max and Marie kicked the bucket
- I will look these student numbers up
- I love them all, kith and kin alike
- My bad for adding fuel on the fire

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Can you identify some properties shared by all these MWEs ?

MWE properties

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No clear consensus, but some shared properties

1. A MWE has several components
2. A MWE is somewhat fixed
3. A MWE has a somewhat idiomatic meaning

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Parameters: the things we use to study a MWE.
Properties: the rules supposedly shared among all MWEs.

1. A MWE has several components

A MWE contains several units

- In latin languages, 1 unit ≈ 1 word
- Word: a sequence of characters delimited by spaces

[3] those who are in the houses
[4] killing two birds with one stone

1. A MWE has several components

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Limits

- *Hot Dog* (english): two words ?
- *Evlerimdekiler*[3] (turkish): one word ?
- *faire d'une pierre deux coups*[4] (french): several words ?

[3] those who are in the houses
[4] killing two birds with one stone

1. A MWE has several components

Max	kicked	the	bucket
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2. A MWE is somewhat fixed

The components of a MWE are stables

- *Fish and Chips*
- *May the force be with you*
- *I will kill / killed two birds with one stone*

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Limits

- Some MWEs can accept slight variations
- Some MWEs can accept word substitutions

2. A MWE is somewhat fixed

Variation	**MWE**	**valid ?**
Base	Max kicked the bucket	yes
Tense	Max is going to kick the bucket	yes
Adverb	Max will kick the bucket in a few hours	yes
Insertion	Max has unfortunatly kicked the bucket	yes
Passivation	The bucket of Max was kicked	no
Clivage	This is the bucket that Max kicked	no
Substitution	Max kicked the vessel / pot	no

Adapted from Gross (1982).

3. A MWE has a somewhat idiomatic meaning

MWEs tend to be non-compositional

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- MWE Meaning $\neq$ sum of its components

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Limits

- Semi-compositional MWEs (*adding fuel on the fire*)
- Fully-compositional MWEs (*to make a decision*)

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Semantic opacity of a MWE

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Reality: it's a *continuum*.

One last property: the most important one.

Take your guesses !

4. A MWE is Conventionalized

Conventionality

- Conventionalized MWE: known by a group of individuals
- **Conventionality is a trait shared by all MWEs**

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About the notion of "group of individuals"

- The population of a region, a state, a country, etc
- The individuals inside this class

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Let's start with a question.

Do you think it is useful to process MWEs in NLP ?

MWEs in NLP

(Some) Interests

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- Better contextualization
  - Jackendoff (1997): 50 % of MWEs in english texts
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  - Does LLMs use fewer MWEs than humans ?

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- (Much) better translations
  - hang on ! ⇒ pends-toi
  - it's raining cats and dogs ⇒ il pleut des chats et des chiens

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Two MWE-related tasks in NLP: discovery and identification.

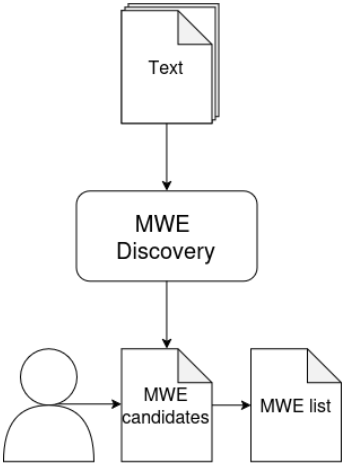
MWE Discovery

Goals

- Discover MWEs from texts
- No prior knowledge on MWEs

Methods

- N-gram frequency analysis
- Manual annotation of all candidates



MWE Discovery with Association Measure (Church and Hanks, 1989)

$$I(x,y) = log_2 \frac{P(x,y)}{P(x)P(y)}$$

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$$I(x,y) = log_2 \frac{P(x,y)}{P(x)P(y)}$$

computer	scientist	1.12
keeping	from	2.14
set	up	7.30

a	doctor	0.96
set	about	-0.60
man	women	-0.12

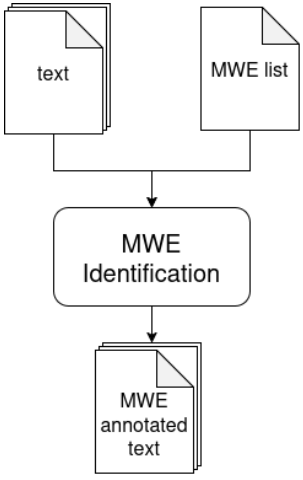
MWE Identification

Goals

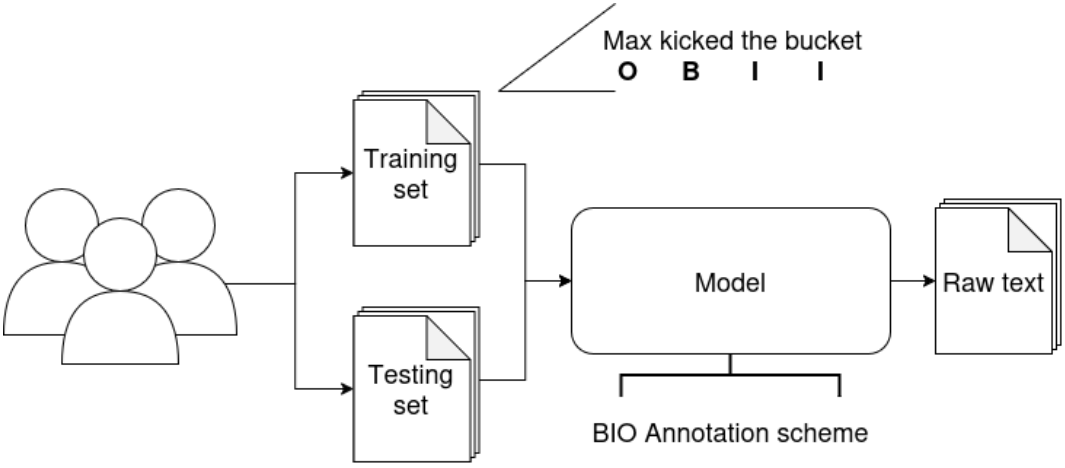
- Annotate MWEs in texts
- Take variation into account

Methods

- Rule-based or statistical
- Based on a list of expected MWEs



MWE Identification with Statistical Approaches



Exercise Time !

What about rule-based MWE identification ?

Find some rules which might help identificating MWEs in texts.

MWE Identification with Rule-based Approaches

Seen2Seen set of rules (Pasquer et al., 2020)

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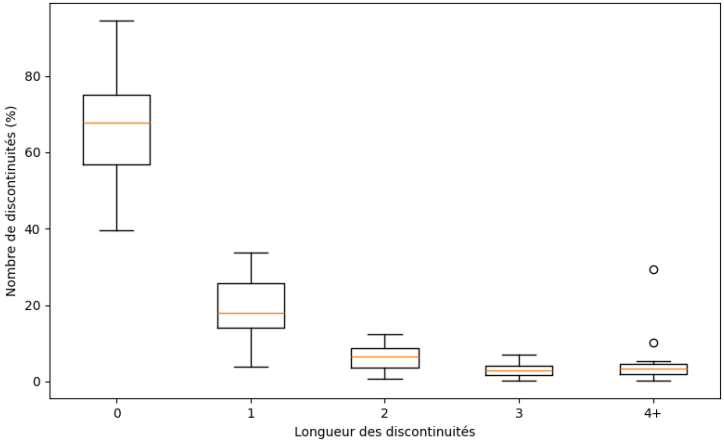
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- Use POS tags of each word
- Consider discontinuities (maximum distance between units ?)
- Always favor closest components
- Units must have syntactic connections
- Consider lexical variations
- Consider nested MWEs (within each others)

About Discontinuities



Doggy bag

Multiword Expressions

- Studied in phraseology
- Several components
- Somewhat fixed
- Somewhat idiomatic
- Always conventionalized

MWEs and NLP

- MWE discovery
- MWE identification



References I

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